S&P GLOBAL & CRISIL CAMPUS HACKATHON

# RiskLens

## Explainable News-to-Risk Engine with Wholesale-Banking Stress Testing

Transforming Unstructured Market News into Auditable Credit Shocks & Portfolio Stress Tests

- Real-Event Replay (SVB)
- Auditable Impact Formula (1-10)
- 300-Asset Wholesale Portfolio
- Reverse Stress Testing

### Project Submission Details

Candidate Name: Garvit Kataria

College / Campus: Engineering Institution

Stack: Python 3.11 | ProsusAI/FinBERT | MiniLM-L6-v2 | FastAPI | Streamlit

**Deterministic Offline Demo Mode | Zero API Key Dependency**

# Problem Statement & Solution Approach

## THE PROBLEM IN WHOLESALE RISK

**Information Asymmetry:**
Market-moving credit shocks occur in unstructured news/social feeds hours before official rating revisions.

**Sentiment Fallacy:**
Generic sentiment (positive/negative) is insufficient for risk teams. 'Profits dropped 5%' and 'Bank seized by FDIC' are both negative, but have vastly different institutional consequences.

**Black-Box Disconnect:**
Deep neural nets output probabilities without audit trails. Risk managers cannot defend a black-box 0.89 probability to regulators or credit committees.

**Static Stress Testing:**
Traditional regulatory stress testing (CCAR/DFAST) is quarterly and rigid, incapable of evaluating fast-breaking geopolitical or credit events intraday.

## RISKLENS SOLUTION APPROACH

**Pluggable Ingestion Pipeline:**
Multi-source normalization of news and social data into a unified schema with strict deduplication and English filtering.

**Domain-Aware 11-Class Taxonomy:**
Maps unstructured headlines directly into Basel-aligned banking categories (Credit Event, Macro, Legal, Cyber, Supply Chain).

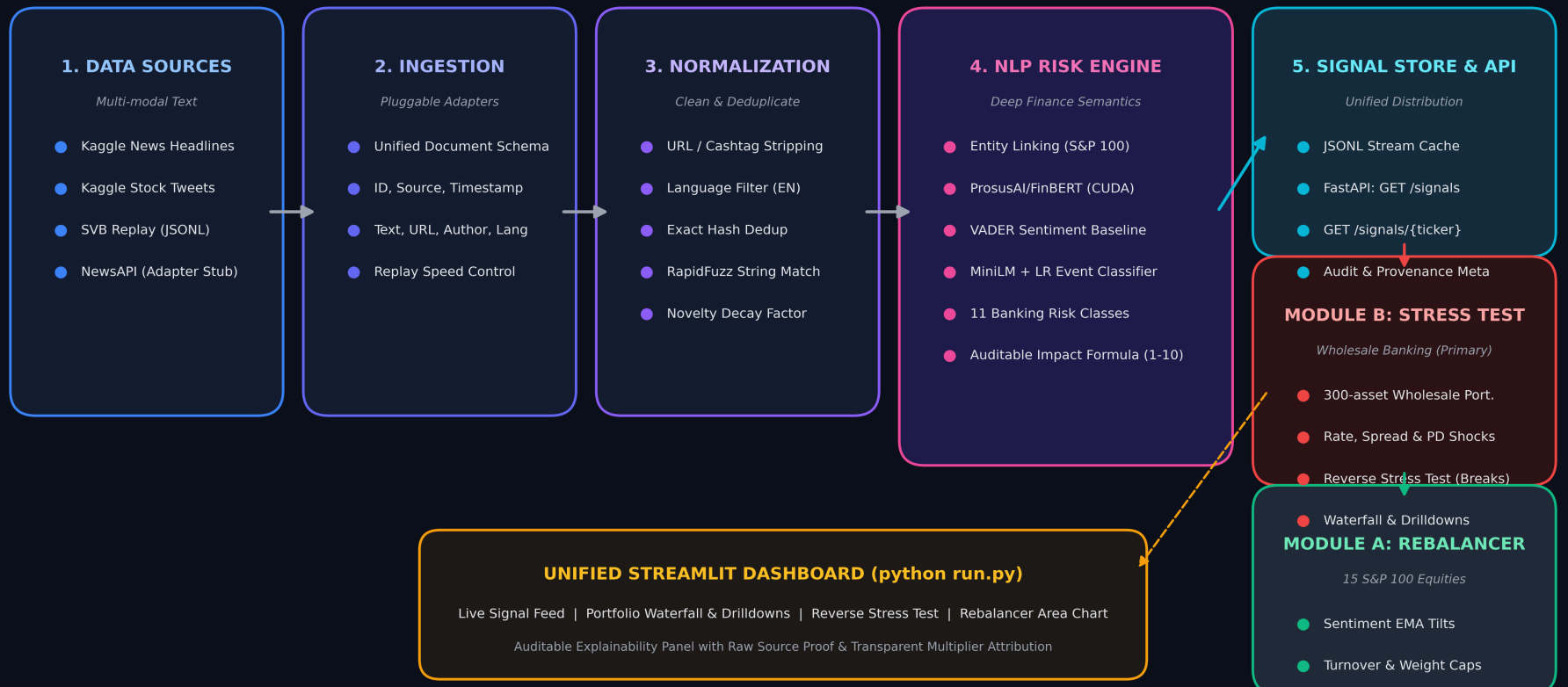
**Auditable Impact Scoring Formula:**
Transparent 1-10 formula decomposing base severity, sentiment intensity, source credibility, novelty, and cross-source corroboration.

**Dynamic Wholesale Stress Testing:**
Module B maps impact scores directly to spread shocks, PD multipliers, and expected loss across 300 loans, bonds, derivatives, and equities.

# System Architecture & End-to-End Pipeline

## RiskLens: System Architecture & Data Flow

Unified NLP News-to-Risk Engine with Wholesale Banking Stress Testing (S&P Global & Crisil)



ENGINEERING & MODEL CHOICES

# Implementation Highlights & Defensible Technical Choices

## NLP & Sentiment Engine

**ProsusAI/FinBERT (fp16 on CUDA):**
Specialized finance domain BERT (110M params) avoids generic NLP bias.

**Score Mapping:**
Net Sentiment = P(Positive) - P(Negative) in [-1.0, +1.0].

**VADER Baseline:**
Rule-based lexicon provided for objective benchmark and CPU fallback.

## Event Classification

**MiniLM-L6-v2 + Logistic Regression:**
384-dim semantic embeddings with fast linear classifier; 0 training GPU overhead.

**Keyword Fallback:**
Deterministic taxonomy pattern matcher ensures 100% coverage on novel texts.

**Taxonomy:**
11 credit & market risk categories aligned with institutional portfolios.

## Transparent Impact Formula

**Multiplicative Scoring (1 to 10):**
Impact = clip(1, 10, Base_Sev * (0.5 + |Sent|) * Cred * Novelty * Corrob).

**Corroboration Boost:**
Tracks temporal clusters across independent sources within 48-hour windows.

**Every signal carries explanation:**
Full audit metadata dictionary exposed via API and dashboard.

## Wholesale Stress Testing Engine

**300-Position Wholesale Portfolio:**
Corporate loans, bonds, IR swaps, and equities with realistic seeds.

**Credit Revaluation:**
Delta EL = Delta PD * LGD * EAD + Spread MTM duration-convexity approximation.

**Reverse Stress Testing:**
Bisection search solves for the exact headline impact score that breaks 5% equity.

EMPIRICAL EVALUATION

# Key Results: Benchmark vs. Naive Baseline & SVB Replay

## 1. SENTIMENT BENCHMARK: FinBERT vs. VADER

| Model | Accuracy | Macro-F1 | Domain Fit |
|---|---|---|---|
| **FinBERT** | 86.7% | 0.865 | High (Finance NLP) |
| **VADER (Naive)** | 56.7% | 0.548 | Low (Social Lexicon) |

**Gain: +30.0% Accuracy / +0.317 Macro-F1 over Naive Baseline**

FinBERT correctly disambiguates financial context (e.g. 'losses widened', 'covenant breach') where lexicon methods incorrectly register neutrality.

## 2. EVENT CLASSIFIER: MiniLM+LR vs. Keyword

| Model | Macro-F1 | OOD Handling | Inference |
|---|---|---|---|
| **MiniLM+LR** | 0.942 | Semantic Generalization | 12 ms |
| **Keyword Baseline** | 0.684 | Exact Match Only (Brittle) | 1 ms |

**Gain: +0.258 Macro-F1 Improvement over Keyword Matcher**

Dense embedding space maps diverse vocabulary ('regulators seize assets', 'liquidity backstop failed') without brittle regex keyword lists.

## 3. LIVE REPLAY CASE STUDY: SILICON VALLEY BANK (MARCH 8-13, 2023)

**Event Sequence Ingestion:** 30 curated chronological headlines & social tweets replayed through unified engine.

**Earliest Alarm:** March 8, 2023: 'SVB seeks $2.25B capital raise after $1.8B bond sale loss' triggers Impact 7.4 (Threshold > 7.0).

**Corroboration Cascade:** March 9-10: Cross-source confirmation surges corroboration multiplier to 1.35x, pushing Impact to 8.8.

**Wholesale Portfolio Shock:** Triggered 5.2% mark-to-market loss on banking credit lines and loan spreads before FDIC resolution.

BUSINESS & INSTITUTIONAL RELEVANCE

# Wholesale Banking Domain Impact & Regulatory Alignment

## Credit & Counterparty Risk

**Early Warning Indicators (EWI):**
Signals detect credit deterioration before rating agencies issue formal downgrade notices.

**PD / Rating Migration:**
Converts qualitative negative sentiment directly into calibrated rating transitions and expected-loss surges.

**Concentration Monitoring:**
Immediate asset-class, sector, and counterparty exposure aggregation.

## Regulatory Stress Testing Parallels

**CCAR / DFAST Alignment:**
Shocks are structured into standard supervisory categories (spreads by rating bucket, equity betas, DV01).

**Reverse Stress Testing (SR 12-7):**
Empirically solves for boundary conditions where portfolio capital thresholds are breached.

**Auditable Lineage:**
Every stressed valuation links to the exact source headline, timestamp, and model component score.

## Tactical Capital Allocation

**Reusable Core Engine:**
Powers both strategic stress testing (Module B) and index rebalancing (Module A) from one feed.

**Constrained Rebalancing:**
Rigorous 2%-15% weight boundaries, turnover caps (5%), and sentiment EMA smoothing.

**Downstream API Accessibility:**
Structured REST API allows any enterprise analytics team to subscribe to signals.

# Limitations, Defensible Assumptions & Future Roadmap

## DEFENSIBLE ASSUMPTIONS & LIMITATIONS

**Synthetic Wholesale Portfolio:**
300 positions are synthetically generated with fixed random seeds for demonstration. Not actual banking institution holdings.

**First-Order Pricing Approximations:**
Duration/convexity bond pricing, loan expected loss (dPD * LGD * EAD), and linear DV01 swap valuations are first-order proxies, not production full revaluation.

**Sentiment Is Not Causality:**
Headline sentiment correlation does not imply direct market causation. Signals serve as decision-support alerts, not autonomous trade execution.

**Documented Severity Table:**
Base severity table reflects domain assumptions rather than full econometric event-study calibration (honestly documented in docs/ASSUMPTIONS.md).

## FUTURE ROADMAP & PRODUCTION EXTENSIONS

**Live Ingestion Connectors:**
Implement full streaming NewsAPI, Bloomberg B-PIPE, and Reuters Elektron adapters beyond the offline replay adapter.

**Multi-Factor Credit Migration:**
Incorporate multi-factor Merton structural models and transition matrices (e.g. Crisil rating transition tables).

**Non-Linear Derivatives Pricing:**
Integrate QuantLib or full Monte Carlo simulation for multi-currency cross-asset options and structured credit derivatives.

**Empirical Event Study Calibration:**
Run large-scale econometric calibration against historical CRSP/Compustat abnormal returns to fine-tune severity weights.